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Dynamic Routes

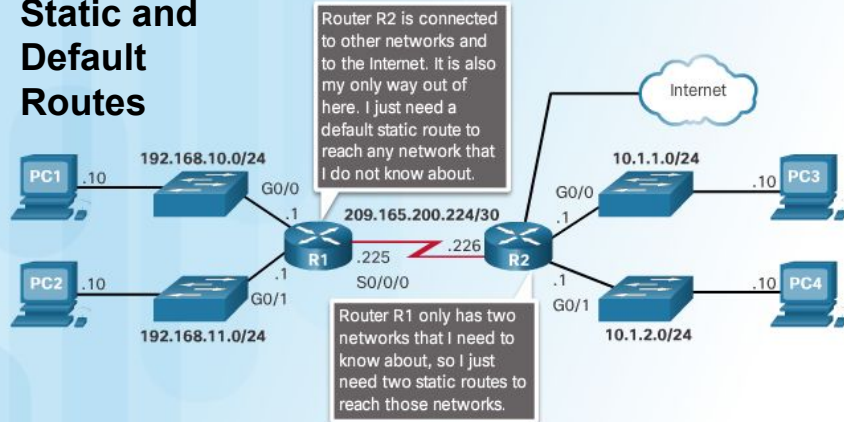
Recall: Static Routing: Reach Remote Networks



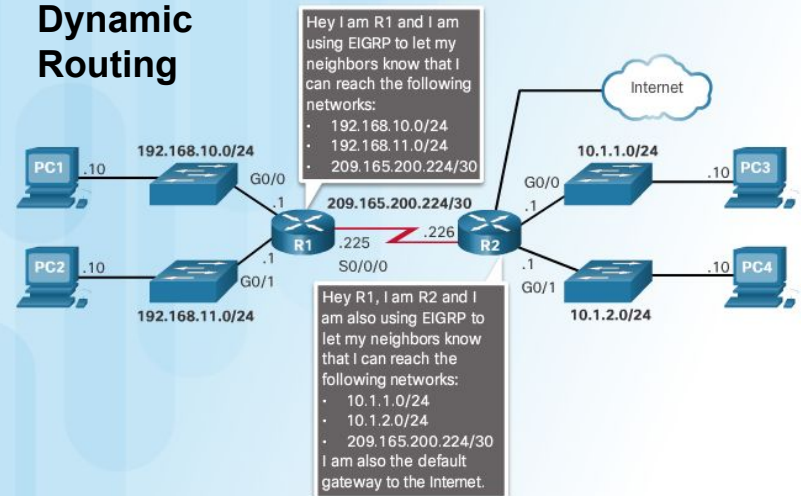
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- A router learns about remote networks in two ways:
 - Manually entered into the route table using static routes
 - Static routes are not automatically updated and must be reconfigured when topology changes
 - Dynamically (Automatically) learned using a routing protocol

Static and Default Routes



Dynamic Routing



Recall: Static Routing: Why Use Static Routing?



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Dynamic versus Static Routing

	Dynamic Routing	Static Routing
Configuration Complexity	Generally independent of the network size	Increases with network size
Topology Changes	Automatically adapts to topology changes	Administrator intervention required
Scaling	Suitable for simple and complex topologies	Suitable for simple topologies
Security	Less secure	More secure
Resource Usage	Uses CPU, memory, link bandwidth	No extra resources needed
Predictability	Route depends on the current topology	Route to destination is always the same

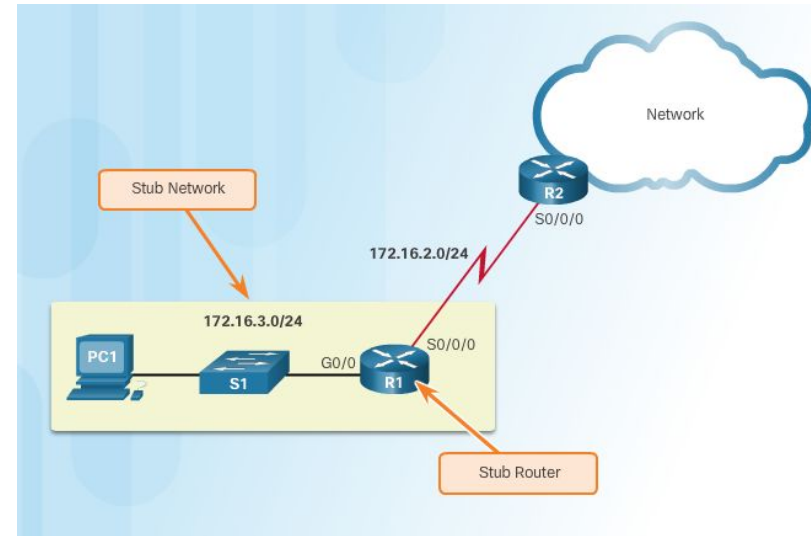
Recall- Static Routing: When to Use Static Routes



Three uses for static routes:

- Smaller networks that are not expected to grow
- Routing to and from stub networks
 - Stub network accessed by a single route and has one neighbor
 - 172.16.3.0 is a stub network
- A single default route to represent a path to any network not found in the routing table
 - Use default route on R1 to point to R2 for all other networks

Stub Networks and Stub Routers





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Dynamic Routing Protocols

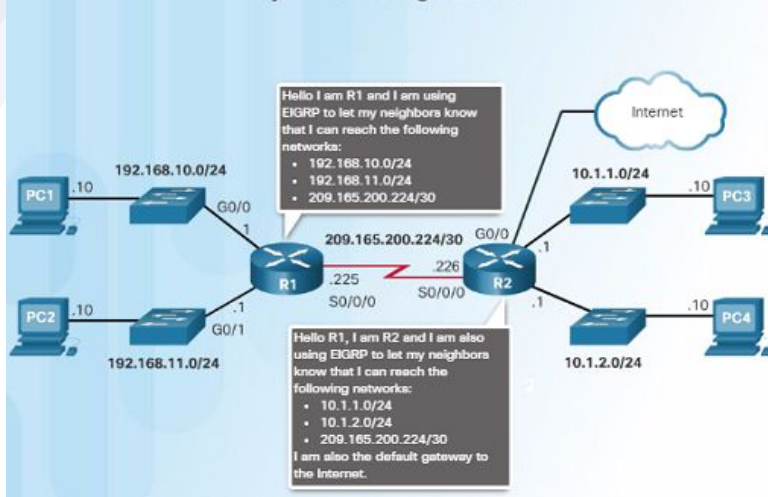
Dynamic Routing Protocols: Dynamic Routing



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- Dynamic routing protocols are used by routers to share information about the reachability and status of remote networks.
- Rather than manually configuring static routes, dynamic routing protocols use network discovery to share information about the networks that it knows about with other routers that are using the same routing protocol.
 - Routers automatically learn about remote networks from other routers
 - These networks and the best path to each are added to the routing table of the router.
- Routers have converged after they have finished exchanging and updating their routing tables. Routers then maintain the networks in their routing tables.

Dynamic Routing Scenario



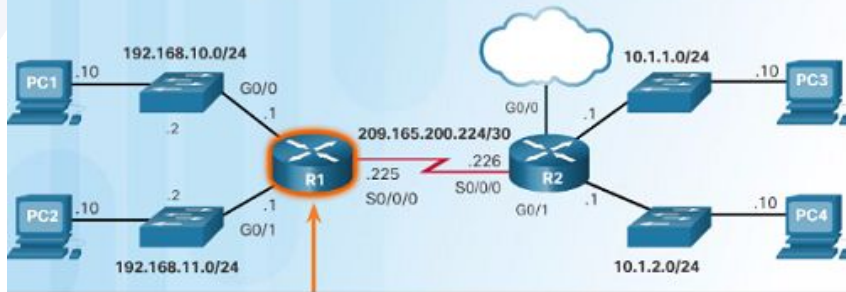
Dynamic Routing Protocols: IPv4 Routing Protocols



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- One of the major advantages of dynamic routing protocols over static routes -determine a new best path if the initial path becomes unusable.
- Dynamic routing protocols can adjust to topology changes without involving the network administrator.
- Cisco routers support a variety of IPv4 routing protocols including:
 - EIGRP
 - OSPF
 - IS-IS
 - RIP
 - Use **router ?** in global config mode to see the complete list.

Supported IPv4 Routing Protocols



```
R1(config)# router ?
bgp      Border Gateway Protocol (BGP)
eigrp    Enhanced Interior Gateway Routing Protocol (EIGRP)
isis     ISO IS-IS
iso-igrp  IGRP for OSI networks
mobile   Mobile routes
odr      On Demand stub Routes
ospf     Open Shortest Path First (OSPF)
ospfv3   OSPFv3
rip      Routing Information Protocol (RIP)

R1(config)# router
```

Dynamic Routing Protocol Overview

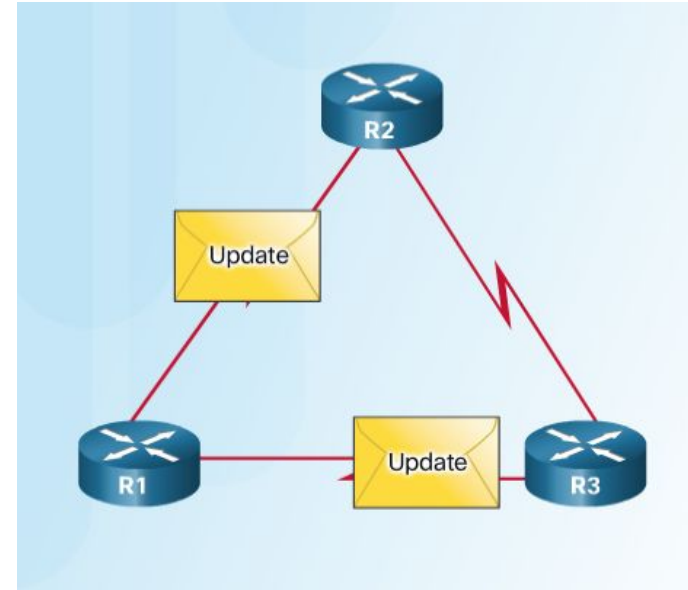
- RIP protocol was updated to RIPv2 to accommodate growth in the network environment
 - RIPv2 does not scale to current larger network implementations
- Routing Protocols developed to meet the need of larger networks include:
 - Open Shortest Path First (OSPF)
 - Intermediate System-to-Intermediate System (IS-IS).
 - Enhanced IGRP (EIGRP)
- Border Gateway Protocol (BGP) is used between Internet service providers (ISPs)

	Interior Gateway Protocols				Exterior Gateway Protocols
	Distance Vector		Link-State		Path Vector
IPv4	RIPv2	EIGRP	OSPFv2	IS-IS	BGP-4
IPv6	RIPng	EIGRP for IPv6	OSPFv3	IS-IS for IPv6	BGP-MP

Dynamic Routing Protocol Components



- Purpose of dynamic routing protocols includes:
 - Discovery of remote networks
 - Maintaining up-to-date routing information
 - Choosing the best path to destination networks
 - Ability to find a new best path if the current path is no longer available
- The main components of dynamic routing protocols include:
 - Data structures - tables or databases kept in RAM.
 - Routing protocol messages - to discover neighboring routers, exchange routing information, and maintain accurate information about the network.
 - Algorithms – to facilitate learning routing information and for best path determination.



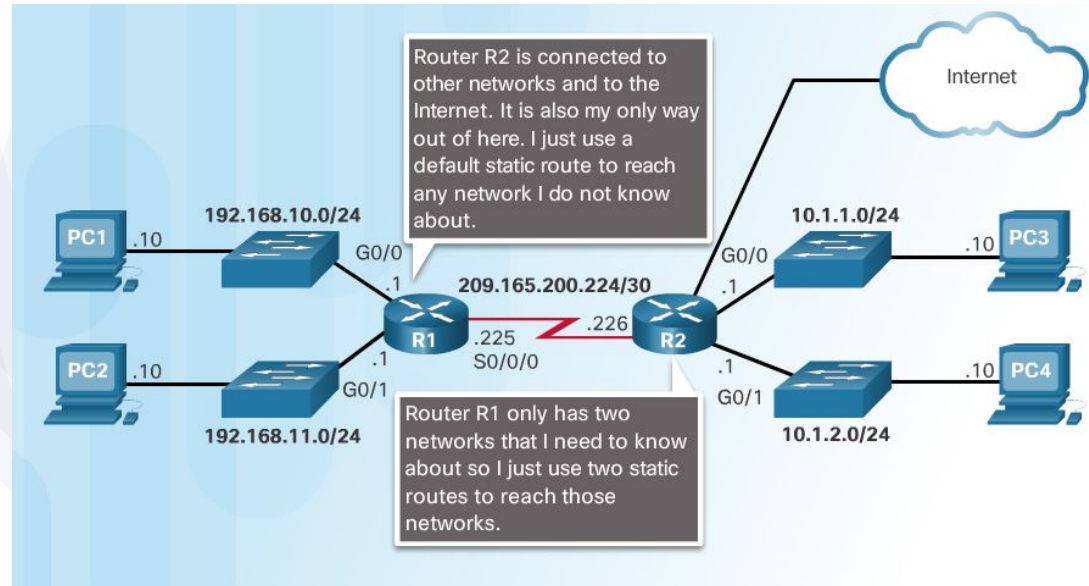
Dynamic versus Static Routing

Static Routing Uses



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- Networks often use both static and dynamic routing.
- Static Routing is used as follows:
 - For easy routing table maintenance in small networks.
 - Routing to and from a stub network.
 - Accessing a single default route.





Advantages	Disadvantages
Easy to implement in a small network.	Suitable only for simple topologies or for special purposes such as a default static route.
Very secure. No advertisements are sent as compared to dynamic routing protocols.	Configuration complexity increases dramatically as network grows.
Route to destination is always the same.	Manual intervention required to re-route traffic.
No routing algorithm or update mechanism required; therefore, extra resources (CPU or RAM) are not required.	

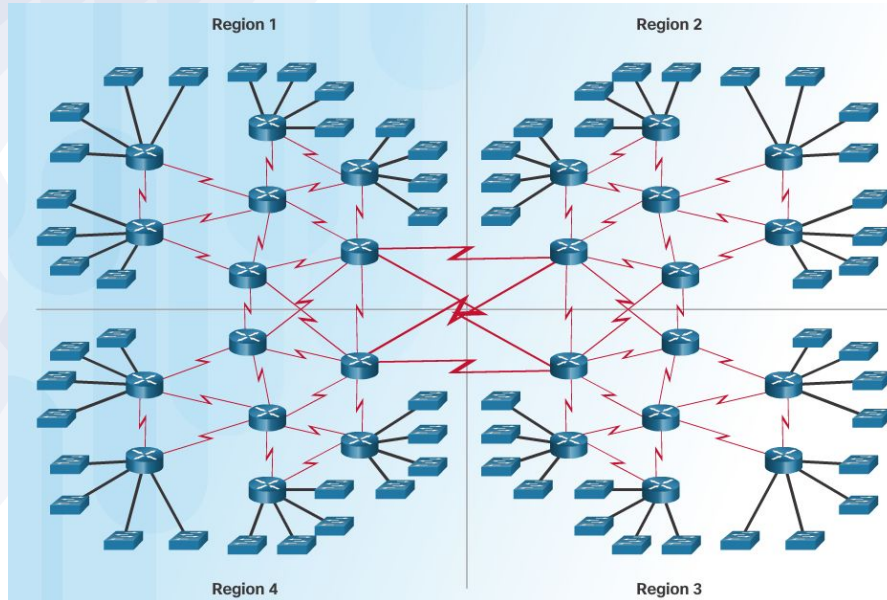
Dynamic versus Static Routing

Dynamic Routing Protocols Uses



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- Dynamic routing is the best choice for large networks
- Dynamic routing protocols help the network administrator manage the network:
 - Providing redundant paths
 - Automatically implementing the alternate path when a link goes down.



Dynamic Routing Advantages and Disadvantages



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Advantages

Suitable in all topologies where multiple routers are required.

Generally independent of the network size.

Automatically adapts topology to reroute traffic if possible.

Disadvantages

Can be more complex to implement.

Less secure. Additional configuration settings are required to secure.

Route depends on the current topology.

Requires additional CPU, RAM, and link bandwidth.



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RIPv2

Routing Information Protocol (RIP)



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- IP RIP (Routing Information Protocol) comes in two different versions: 1 and 2. Version 1 is a distance vector protocol (RFC 1058) and Version 2 is a hybrid protocol (RFCs 1721 and 1722).
- **Routing Information Protocol Version 1:** RIPv1 uses local broadcasts to share routing information. These updates are periodic in nature, occurring, by default, every 30 seconds. To prevent packets from circling around a loop forever, both versions of RIP solve counting to infinity by placing a hop count limit of 15 hops on packets. Any packet that reaches the sixteenth hop will be dropped. RIPv1 is a classful protocol. RIP supports up to six equal-cost paths to a single destination. Equal-cost path are the paths where the metric is same (Hop count).
- **Routing Information Protocol Version 2:** RIPv2 is a distance vector routing protocol with routing enhancements built into it, and it is based on RIPv1. Therefore, it is commonly called as hybrid routing protocol.

Routing Information Protocol (RIP)



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- RIPv2 uses multicasts instead of broadcasts. RIPv2 supports triggered updates. when a change occurs, a RIPv2 router will immediately propagate its routing information to its connected neighbours.
- RIPv2 is a classless protocol and it supports variable-length subnet masking (VLSM). Both RIPv1 and RIPv2 uses hop count as the metric.

Differences between RIPv1 and RIPv2

- RIPv1
 - Supports only classful routing (Does not support VLSM).
 - No authentication.
 - RIPv1 uses Broadcast.

Routing Information Protocol (RIP)



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- RIPv2
 - Supports classless routing (Supports VLSM and CIDR). Incorporates the addition of the network mask in the update to allow classless routing advertisements.
 - Authentication is available.
 - RIPv2 uses multi-cast instead of broadcast. multicast communication reduces the burden on the network devices that do not need to listen to

```
Router>enable
```

```
Router#configure terminal
```

```
Router(config)# router rip
```

```
Router(config-router)# version 2
```

```
Router(config-router)# network network_id
```

```
Router02>enable
Router02#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router02 (config)#router rip
Router02 (config-router)#version 2
Router02 (config-router)#network 172.17.0.0
Router02 (config-router)#network 172.18.0.0
Router02 (config-router)#network 172.19.0.0
Router02 (config-router)#exit
Router02 (config)#exit
Router02#
```

Dynamic versus Static Routing

Router RIP Configuration Mode



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- Use the **router rip** command to enable RIP
- Use the **no router rip** command to disable

```
R1# conf t
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)# router rip
R1(config-router)#
```

RIP Configuration Options

```
R1(config-router)# ?

Router configuration commands:
address-family      Enter Address Family command mode
auto-summary        Enable automatic network number summarization
default             Set a command to its defaults
default-information  Control distribution of default information
default-metric       Set metric of redistributed routes
distance            Define an administrative distance
distribute-list      Filter networks in routing updates
exit                Exit from routing protocol configuration mode
flash-update-threshold Specify flash update threshold in second
help                Description of the interactive help system
input-queue          Specify input queue depth
maximum-paths        Forward packets over multiple paths
neighbor            Specify a neighbor router
network             Enable routing on an IP network
no                  Negate a command or set its defaults
offset-list          Add or subtract offset from RIP metrics
output-delay         Interpacket delay for RIP updates
passive-interface    Suppress routing updates on an interface
redistribute         Redistribute information from another routing protocol
timers              Adjust routing timers
traffic-share         How to compute traffic share over alternate paths
validate-update-source Perform sanity checks against source address of routing updates
version             Set routing protocol version

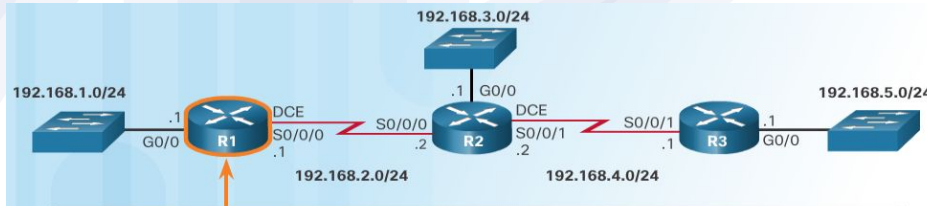
R1(config-router)#
```

Configuring the RIP Protocol Advertise Networks



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- The **network** *network-address* router configuration mode command:
 - Enables RIP on all interfaces that belong to a specific network
 - Advertises the network in RIP routing updates sent to other routers every 30 seconds.



```
R1(config)# router rip
R1(config-router)# network 192.168.1.0
R1(config-router)# network 192.168.2.0
R1(config-router)#
```

Note: RIPv1 is a classful routing protocol for IPv4.

Configuring the RIP Protocol

Verify RIP Routing



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```
R1# show ip protocols
*** IP Routing is NSF aware ***
```

Routing Protocol is "rip"

Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Sending updates every 30 seconds, next due in 16 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Redistributing: rip

Default version control: send version 1, receive any version

Interface	Send	Recv	Triggered RIP	Key-chain
GigabitEthernet0/0	1	1 2		
Serial0/0/0	1	1 2		

Automatic network summarization is in effect:

Maximum path: 4

Routing for Networks:

192.168.1.0
192.168.2.0

Routing Information Sources:

Gateway	Distance	Last Update
192.168.2.2	120	00:00:15

Distance: (default is 120)

R1#

show ip protocols – displays IPv4 routing protocols configured on the router.

```
R1# show ip route | begin Gateway
Gateway of last resort is not set
```

192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.1.0/24 is directly connected,
GigabitEthernet0/0
L 192.168.1.1/32 is directly connected,
GigabitEthernet0/0
192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C 192.168.2.0/24 is directly connected, Serial0/0/0
L 192.168.2.1/32 is directly connected, Serial0/0/0
R 192.168.3.0/24 [120/1] via 192.168.2.2, 00:00:24,
Serial0/0/0
R 192.168.4.0/24 [120/1] via 192.168.2.2, 00:00:24,
Serial0/0/0
R 192.168.5.0/24 [120/2] via 192.168.2.2, 00:00:24,
Serial0/0/0
R1#

show ip route – displays RIP routes installed in the routing table.

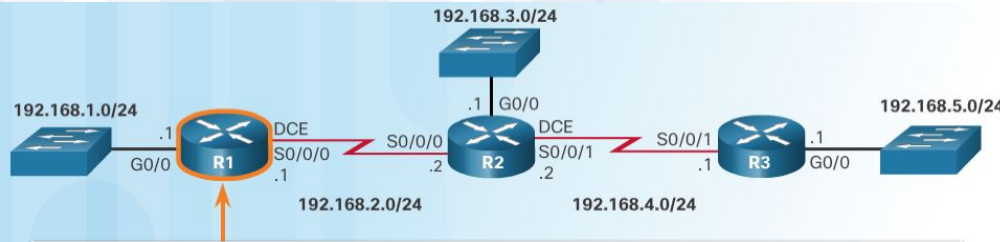
Configuring the RIP Protocol

Enable and Verify RIPv2



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- Use the **version 2** router configuration mode command to enable RIPv2
- Use the **show ip protocols** command to verify that RIPv2 is configured.
- Use the **show ip route** command to verify the RIPv2 routes in the routing table.



```
R1(config)# router rip
R1(config-router)# version 2
R1(config-router)# ^Z
R1#
R1# show ip protocols | section Default
Default version control: send version 2, receive version 2
Interface      Send  Recv  Triggered RIP  Key-chain
GigabitEthernet0/0    2    2
Serial0/0/0          2    2
R1#
```

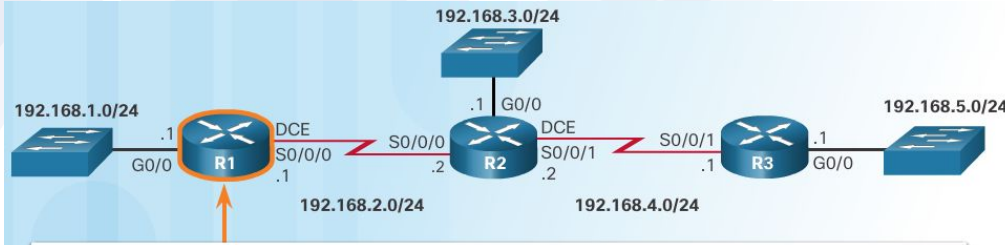
Configuring the RIP Protocol

Disable Auto Summarization



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- RIPv2 automatically summarizes networks at major network boundaries.
- Use the **no auto-summary** router configuration mode command to disable auto summarization.
- Use the **show ip protocols** command to verify that auto summarization is off.



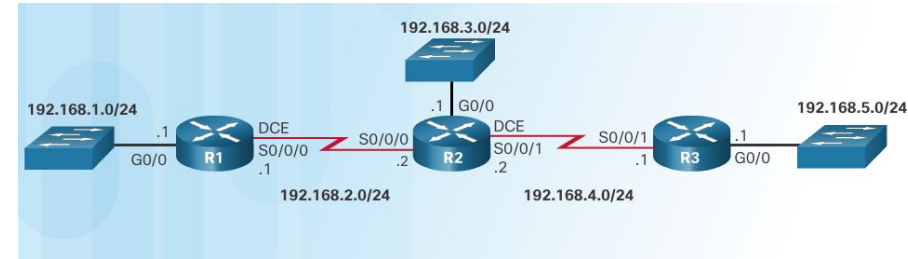
```
R1(config)# router rip
R1(config-router)# no auto-summary
R1(config-router)# end
R1#
*Mar 10 14:11:49.659: %SYS-5-CONFIG_I: Configured from
console by console
R1# show ip protocols | section Automatic
Automatic network summarization is not in effect
R1#
```

Configuring the RIP Protocol

Configure Passive Interfaces



- RIP updates:
 - Are forwarded out all RIP-enabled interfaces by default.
 - Only need to be sent out interfaces that are connected to other RIP-enabled routers.
- Sending RIP updates to LANs wastes bandwidth, wastes resources, and is a security risk.
- Use the **passive-interface** router configuration command to stop routing updates out the interface. Still allows that network to be advertised to other routers.



```
R1(config)# router rip
R1(config-router)# passive-interface g0/0
R1(config-router)# end
R1#
R1# show ip protocols | begin Default
  Default version control: send version 2, receive version 2
  Interface                Send Recv Triggered RIP Key-
chain
  Serial0/0/0                2    2
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
  192.168.1.0
  192.168.2.0
Passive Interface(s):
  GigabitEthernet0/0
Routing Information Sources:
  Gateway      Distance    Last Update
  192.168.2.2    120        00:00:06
Distance: (default is 120)

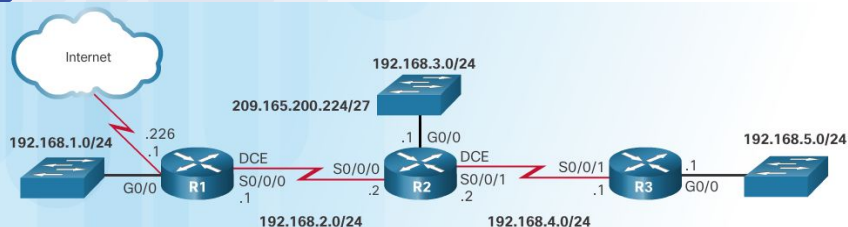
R1#
```

Configuring the RIP Protocol Propagate a Default Route



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- In the diagram a default static route to the Internet is configured on R1.
- The **default-information originate** router configuration command instructs R1 to send the default static route information in the RIP updates.



```
R1(config)# ip route 0.0.0.0 0.0.0.0 S0/0/1 209.165.200.226
R1(config)# router rip
R1(config-router)# default-information originate
R1(config-router)# ^Z
R1#
*Mar 10 23:33:51.801: %SYS-5-CONFIG_I: Configured from console by console
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.226 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 209.165.200.226, Serial0/0/1
    192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.1.0/24 is directly connected, GigabitEthernet0/0
L    192.168.1.1/32 is directly connected, GigabitEthernet0/0
    192.168.2.0/24 is variably subnetted, 2 subnets, 2 masks
C    192.168.2.0/24 is directly connected, Serial0/0/0
L    192.168.2.1/32 is directly connected, Serial0/0/0
R    192.168.3.0/24 [120/1] via 192.168.2.2, 00:00:08, Serial0/0/0
R    192.168.4.0/24 [120/1] via 192.168.2.2, 00:00:08, Serial0/0/0
R    192.168.5.0/24 [120/2] via 192.168.2.2, 00:00:08, Serial0/0/0
    209.165.200.0/24 is variably subnetted, 2 subnets, 2 masks
C    209.165.200.0/24 is directly connected, Serial0/0/1
L    209.165.200.225/27 is directly connected, Serial0/0/1
R1#
```

Dynamic Routing Protocols: RIPng



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- The primary features of RIPng are the same as they were in IPv4: It's still a *distance-vector* protocol, has a **maximum hop** count of **15** and uses **split horizon**, **poison reverse** and other loop avoidance mechanisms.
- RIPng still uses **multicast** to send routing updates, but in IPv6, it uses **FF02::9** for the multicast traffic.
- But there are also some differences between the two versions. One of the most notable changes with RIPng (and all other IPv6 routing protocols), is that you enable network advertisement from interface configuration mode of a router instead of using **network** command in the global configuration mode.

Dynamic Routing Protocols: RIPng



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Router(config-if)#ipv6 rip 10 enable

- The **10** that you see in the command is a number that identifies the **process** of RIPng that's running. Just by typing this command, a new RIPng process is started (enabled) for you so you don't have to go to global configuration mode to enable the process.

```
R1(config)#int fa0/0
```

```
R1(config-if)#ipv6 RIP 10 enable
```

```
R1(config-if)#
```

```
R1(config-if)#int serial 0/0/0
```

```
R1(config-if)#ipv6 RIP 10 enable
```

SUMMARY STEPS

- enable**
- configure terminal**
- ipv6 unicast-routing**
- interface type number**
- ipv6 rip name enable**

Step 1	enable Example: Router#enable	Enables privileged EXEC mode. • Enter your password if prompted.
Step 2	configure terminal Example: Router#configure terminal	Enters global configuration mode.
Step 3	ipv6 router rip word Example: Router(config)#ipv6 router rip process1	Configures an IPv6 RIP routing process and enters router configuration mode for the IPv6 RIP routing process. • Use the word argument to identify a specific IPv6 RIP routing process.
Step 4	maximum-paths number-paths Example: Router(config-router)#maximum-paths 1	(Optional) Defines the maximum number of equal-cost routes that IPv6 RIP can support. • The number-paths argument is an integer from 1 to 64. The default for RIP is four paths.
Step 5	exit Example: Router(config-if)#exit	Exits interface configuration mode and enters global configuration mode.
Step 6	interface type number Example: Router(config)#interface gigabitethernet 0/0/0	Specifies the interface type and number, and enters interface configuration mode.
Step 7	ipv6 rip name default-information [only] originate [metric metric-value] Example: Router(config-if)#ipv6 rip process1 default-information originate	(Optional) Originates the IPv6 default route (::/0) into the specified RIP routing process updates sent out of the specified interface. Note To avoid routing loops after the IPv6 default route (::/0) is originated out of any interface, the routing process ignores all default routes received on any interface. • Specifying the only keyword originates the default route (::/0) but suppresses all other routes in the updates sent on this interface. • Specifying the originate keyword originates the default route (::/0) in addition to all other routes in the updates sent on this interface.

Route Summarization

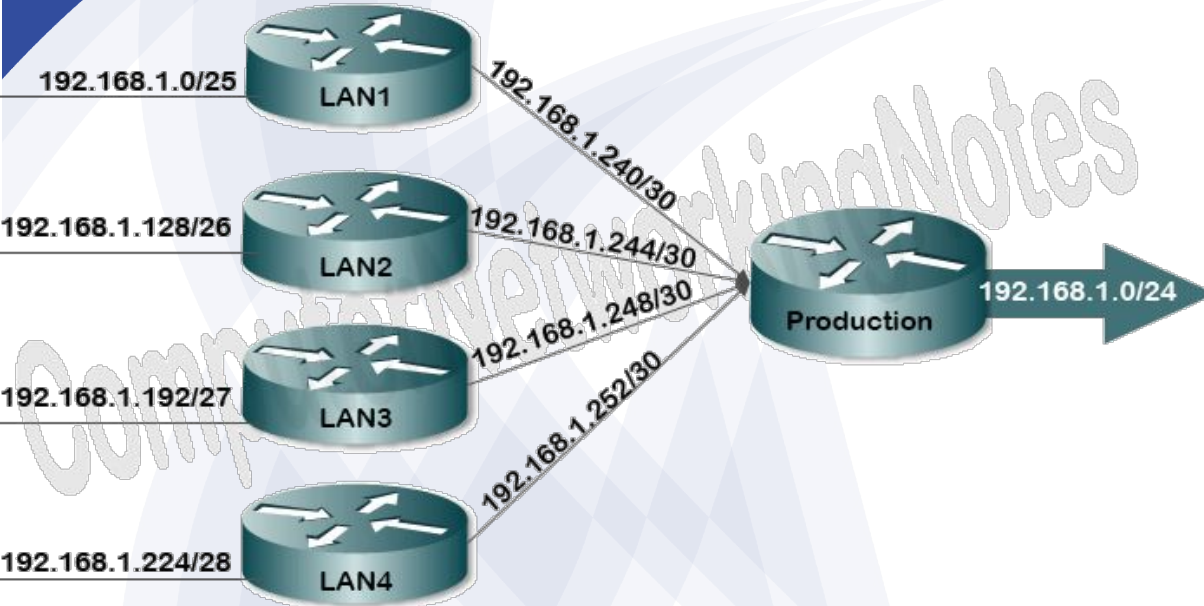


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Summary of
subnetworks

Disadvantages

- It adds complexity to the network.
- Summarized routes are confusing. You can decode them only if you know how subnetting works.
- Summarized routes may contain errors. For example, they may advertise a down subnet as an up subnet



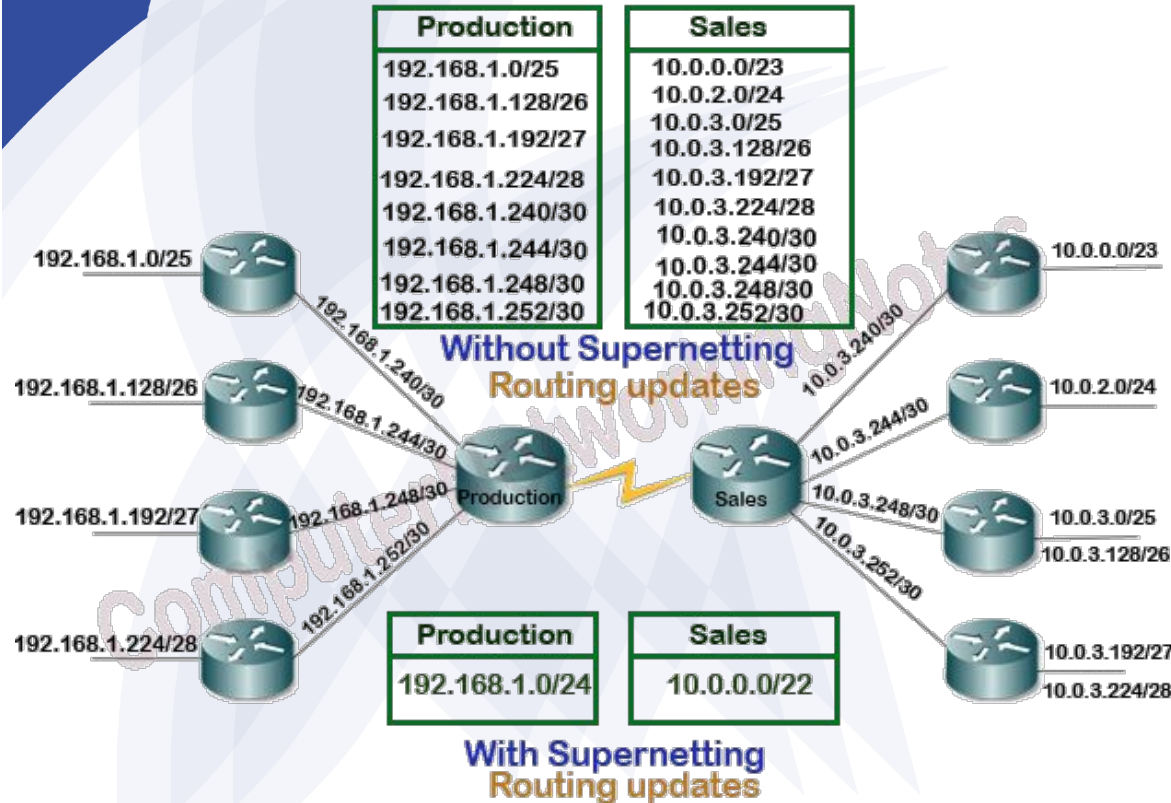
Route Summarization



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Summary of subnetworks

- It reduces the size of routing tables. Routers store only summarized routes in the routing table instead of subnetted.
- Routers need less memory and processing power as they process fewer routes.
- It requires less bandwidth as routers advertise fewer routing updates.
- It hides network problems such as a flapping route.
- It provides a better overview of the network.
- It decreases the use of resources such as Memory and CPU.
- It decreases the required time to rebuild the routing tables.



Route Summarization and Flapping Routes



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- A flapping route is a route that is going up and down, up and down. Loose connections and short circuits are two common reasons that cause flapping.
- A flapping route affects the connected router and any other router that knows about it. Every time it goes up or down, the other routers have to incorporate the change into their routing tables.
- However, if routers know only the summarized route, they will not be affected by the flapping route. They will affect only when all four subnets fail, causing the router performing the summarization to stop advertising the summarized route.
- Hiding a flapping route has both advantages and disadvantages. On the positive side, it provides stability in the network. Routers change their routing table only when all routes of a summarized route fail. On the negative side, it hides the complete picture of the network.
- It can cause routers to make bad assumptions. Let us understand it through the preceding example.
- Suppose, the network 192.168.1.128/26 is down. Routers on the right side of the production router will still receive updates concerning the summarized route (192.168.1.0/24). From their perspective, since the production router is still advertising this route, all addresses from 192.168.1.0 through 192.168.1.255 must be available. Obviously, this is not true, and thus other routers will still send traffic to 192.168.1.128/26 since they think it's still reachable.



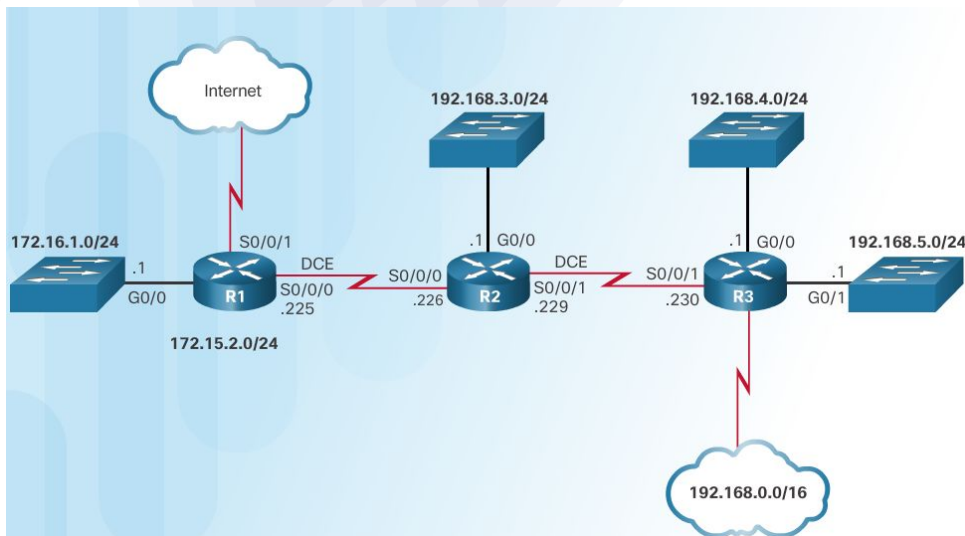
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The Routing Table

Parts of an IPv4 Route Entry



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```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S* 0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
    is directly connected, Serial0/0/1
C 172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
L 172.16.1.0/24 is directly connected, GigabitEthernet0/0
L 172.16.1.1/32 is directly connected, GigabitEthernet0/0
R 172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R 172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R 172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R 192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
    209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C 209.165.200.224/30 is directly connected, Serial0/0/0
L 209.165.200.225/32 is directly connected, Serial0/0/0
R 209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C 209.165.200.232/30 is directly connected, Serial0/0/1
L 209.165.200.233/30 is directly connected, Serial0/0/1
R1#
```

Routing Table for R1

Parts of an IPv4 Route Entry

Directly Connected Entries



- Directly Connected Networks (C) are automatically added to the routing table when the interface is configured and activated.
- Entries contain the following information:
 - Route source - how the route was learned.
 - Destination network – remote network.
 - Outgoing Interface – exit interface used to forward packets to destination.
- Other route source entries include:
 - S –Static Route
 - D – EIGRP routing protocol
 - O – OSPF routing protocol
 - R - RIP routing protocol

Route Source	Destination Network	Outgoing Interface
C	172.16.1.0/24 is directly connected,	GigabitEthernet0/0
L	172.16.1.1/32 is directly connected,	GigabitEthernet0/0

Legend

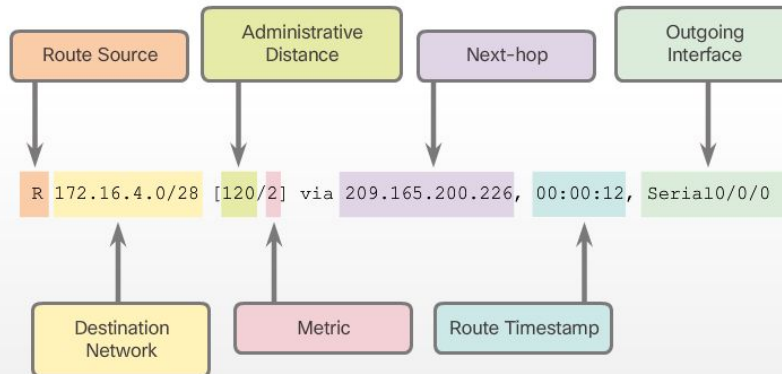
-  - Identifies how the network was learned by the router.
-  - Identifies the destination network and how it is connected.
-  - Identifies the interface on the router connected to the destination network.

Parts of an IPv4 Route Entry

Remote Network Entries



- Routes to remote networks contain the following information:
 - Route source – how route was learned
 - Destination network
 - Administrative distance (AD) - trustworthiness of the route.
 - Metric – value assigned to reach the remote network. Lower is better.
 - Next hop – IPv4 address of the next router that the packet should be forwarded to.
 - Route timestamp – time since the route was updated.
 - Outgoing interface - the exit interface to use to forward the packet



Dynamically Learned IPv4 Routes

Routing Table Terms



- The routing table is a hierarchical structure that is used to speed up the lookup process when locating routes and forwarding packets.
- The hierarchy includes:
 - Ultimate Routes
 - Level 1 routes
 - Level 1 parent routes
 - Level 2 child routes

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*    0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
       is directly connected, Serial0/0/1
      172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C      172.16.1.0/24 is directly connected, GigabitEthernet0/0
L      172.16.1.1/32 is directly connected, GigabitEthernet0/0
R      172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R      172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R      172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R      192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
      209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C      209.165.200.224/30 is directly connected, Serial0/0/0
L      209.165.200.225/32 is directly connected, Serial0/0/0
R      209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C      209.165.200.232/30 is directly connected, Serial0/0/1
L      209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```

Dynamically Learned IPv4 Routes

Ultimate Route



- An ultimate route is a routing table entry that contains either a next-hop IPv4 address or an exit interface.
- Directly connected, dynamically learned, and local routes are ultimate routes.

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
      172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
      C   172.16.1.0/24 is directly connected, GigabitEthernet0/0
      L   172.16.1.1/32 is directly connected, GigabitEthernet0/0
      R   172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
      R   172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
      R   172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
      R   192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
      209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
      C   209.165.200.224/30 is directly connected, Serial0/0/0
      L   209.165.200.225/32 is directly connected, Serial0/0/0
      R   209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
      C   209.165.200.232/30 is directly connected, Serial0/0/1
      L   209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```

Dynamically Learned IPv4 Routes

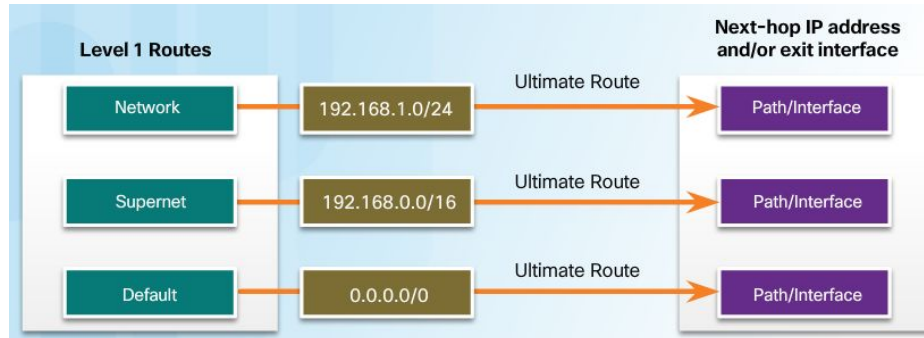
Level 1 Route



- A level 1 route can be a:
 - **Network route** - a network route that has a subnet mask equal to that of the classful mask.
 - **Supernet route** - a network address with a mask less than the classful mask, for example, a summary address.
 - **Default route** - a static route with the address 0.0.0.0/0

```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
C    172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
L    172.16.1.0/24 is directly connected, GigabitEthernet0/0
L    172.16.1.1/32 is directly connected, GigabitEthernet0/0
R    172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R    172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R    172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R    192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
C    209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
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C    209.165.200.232/30 is directly connected, Serial0/0/1
L    209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```



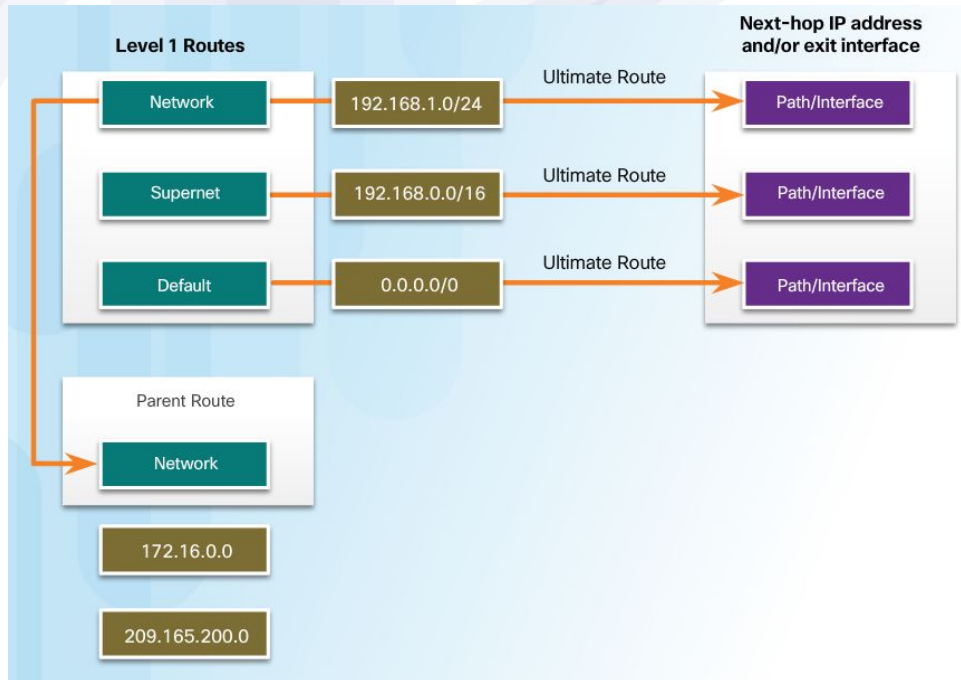
Dynamically Learned IPv4 Routes

Level 1 Parent Route



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- A parent route is a level 1 network route that is subnetted.
- In the routing table, it basically provides a heading for the specific subnets it contains.



```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
      172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
C     172.16.1.0/24 is directly connected, GigabitEthernet0/0
L     172.16.1.1/32 is directly connected, GigabitEthernet0/0
R     172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R     192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
      209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C     209.165.200.224/30 is directly connected, Serial0/0/0
L     209.165.200.225/32 is directly connected, Serial0/0/0
R     209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C     209.165.200.232/30 is directly connected, Serial0/0/1
L     209.165.200.233/32 is directly connected, Serial0/0/1
R1#
```

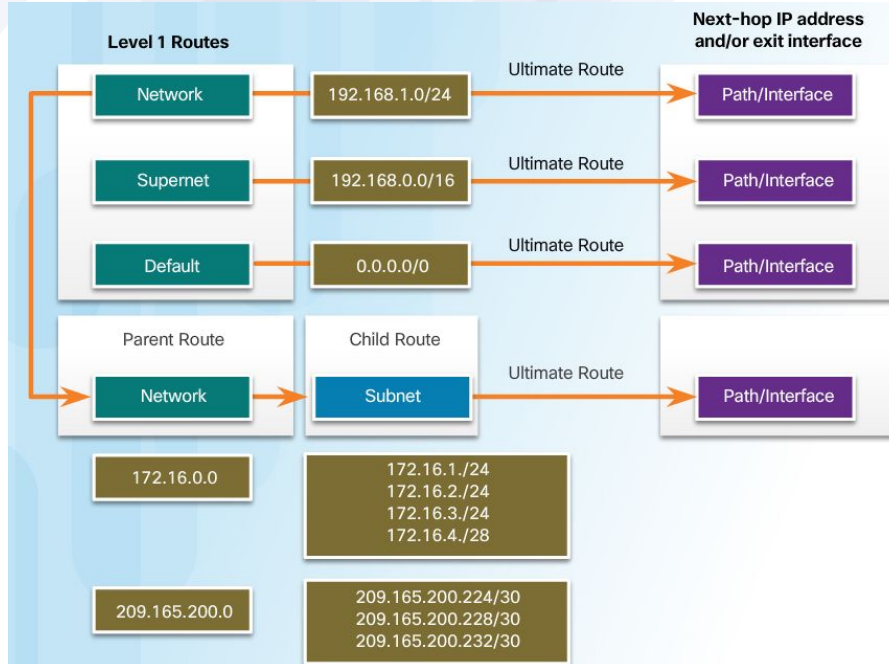
Dynamically Learned IPv4 Routes

Level 2 Child Route



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- A level 2 child route is a route that is a subnet of a classful network address.
- Level 1 parent routes contain level 2 child routes.
- Level 2 child routes are also ultimate routes.



```
R1# show ip route | begin Gateway
Gateway of last resort is 209.165.200.234 to network 0.0.0.0

S*   0.0.0.0/0 [1/0] via 209.165.200.234, Serial0/0/1
      is directly connected, Serial0/0/1
C    172.16.0.0/16 is variably subnetted, 5 subnets, 3 masks
L    172.16.1.0/24 is directly connected, GigabitEthernet0/0
R    172.16.1.1/32 is directly connected, GigabitEthernet0/0
R    172.16.2.0/24 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
R    172.16.3.0/24 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R    172.16.4.0/28 [120/2] via 209.165.200.226, 00:00:12, Serial0/0/0
R    192.168.0.0/16 [120/2] via 209.165.200.226, 00:00:03, Serial0/0/0
C    209.165.200.0/24 is variably subnetted, 5 subnets, 2 masks
C    209.165.200.224/30 is directly connected, Serial0/0/0
L    209.165.200.225/32 is directly connected, Serial0/0/0
R    209.165.200.228/30 [120/1] via 209.165.200.226, 00:00:12, Serial0/0/0
C    209.165.200.232/30 is directly connected, Serial0/0/1
L    209.165.200.233/30 is directly connected, Serial0/0/1
R1#
```

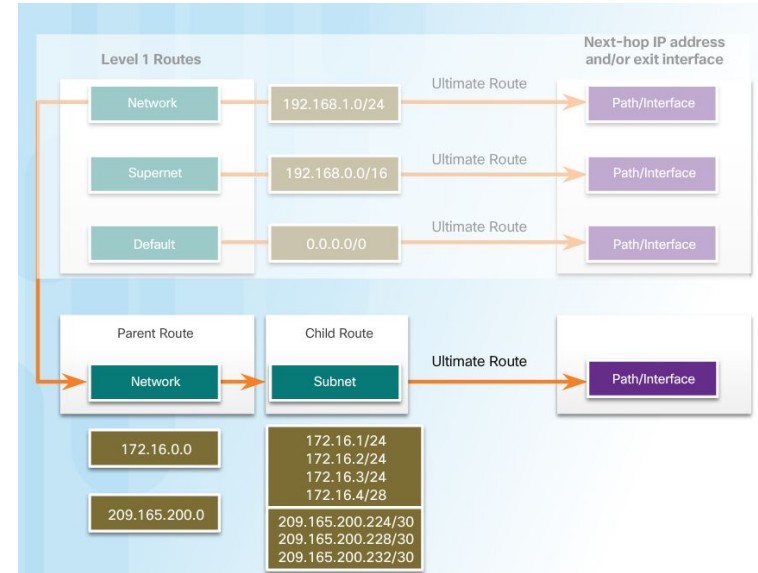
The IPv4 Route Lookup Process

Route Lookup Process



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- Router lookup process:
 - If the best match is a level 1 ultimate route, then this route is used to forward the packet.
 - If the best match is a level 1 parent route, the router then examines child routes (the subnet routes).
 - If there is a match with a level 2 child route, that is used to forward the packet.
 - If there is no match with level 2 child routes, the router searches level 1 supernet or default routes. If there is a match, that route is used.
 - If there is no match found in the routing table the packet is dropped.



The IPv4 Route Lookup Process

Best Route = Longest Match



- The best match is the route in the routing table that has the greatest number of far-left matching bits with the destination IPv4 address of the packet.
- The route with the greatest number of equivalent far left bits, or the longest match, is always the preferred route.

IP Packet Destination	172.16.0.10	10101100.00010000.00000000.00001010
Route 1	172.16.0.0/12	10101100.00010000.00000000.00000000
Route 2	172.16.0.0/18	10101100.00010000.00000000.00000000
Route 3	172.16.0.0/26	10101100.00010000.00000000.00000000

Longest Match to IP Packet Destination

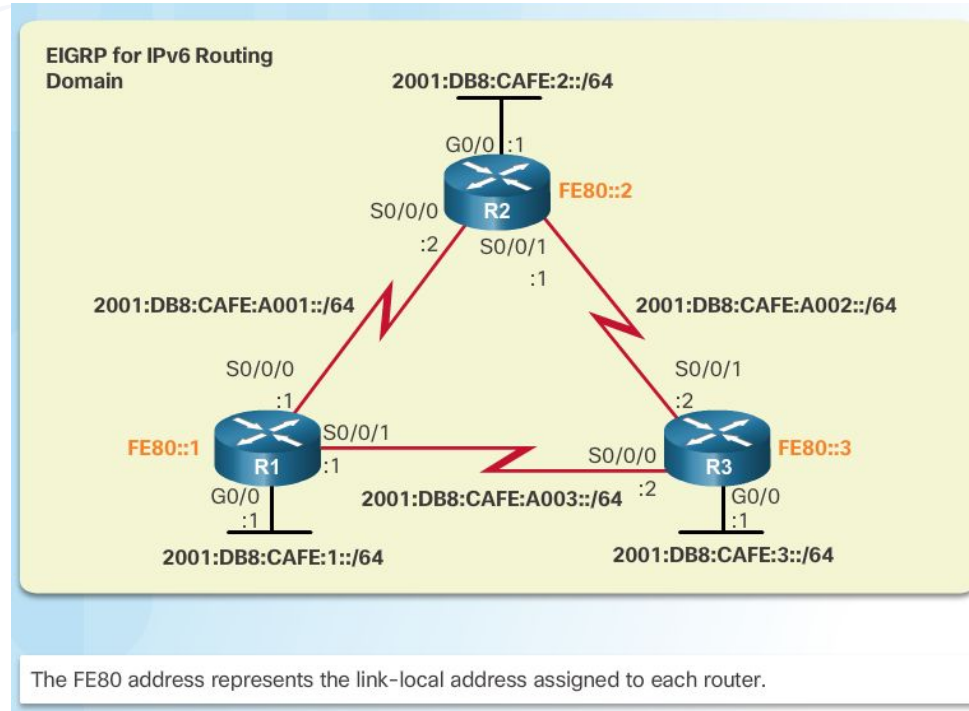
Analyze an IPv6 Routing Table

IPv6 Routing Table Entries



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- An IPv6 routing table includes directly connected, static and dynamically learned routes.
- All IPv6 routes are level 1 ultimate routes.



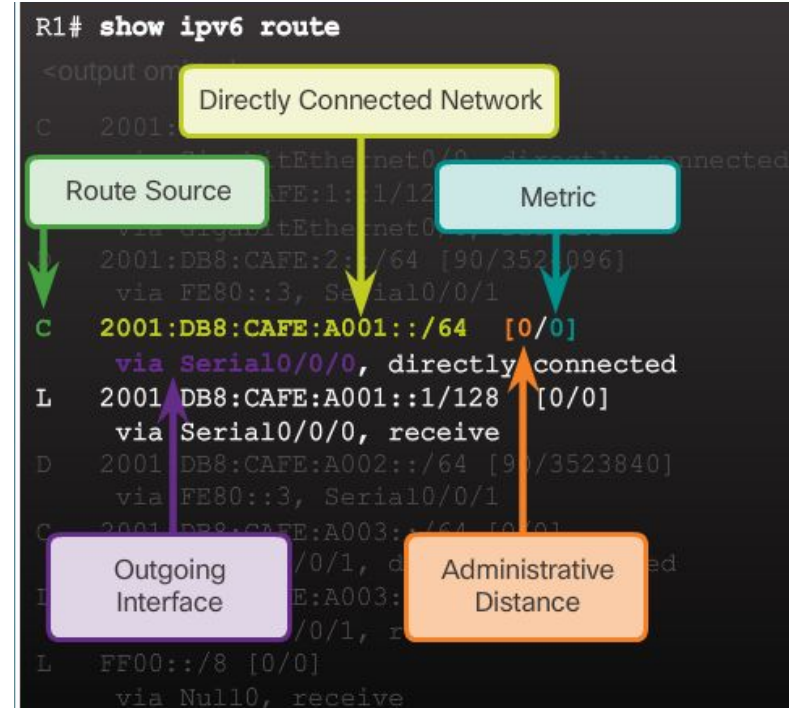
Analyze an IPv6 Routing Table

Directly Connected Entries



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- Use the **show ipv6 route** command to display the IPv6 routing table.
- The directly connected route entries include the following:
 - Route source – How the route was learned. Directly connected indicated with a C and L for local route.
 - Directly connected network address.
 - Administrative distance – Trustworthiness of the route (lower more trustworthy).
 - Metric – Value assigned to reach the network (lower is preferred route).
 - Outgoing interface – Exit interface used to forward packet.



Analyze an IPv6 Routing Table

Remote IPv6 Network Entries

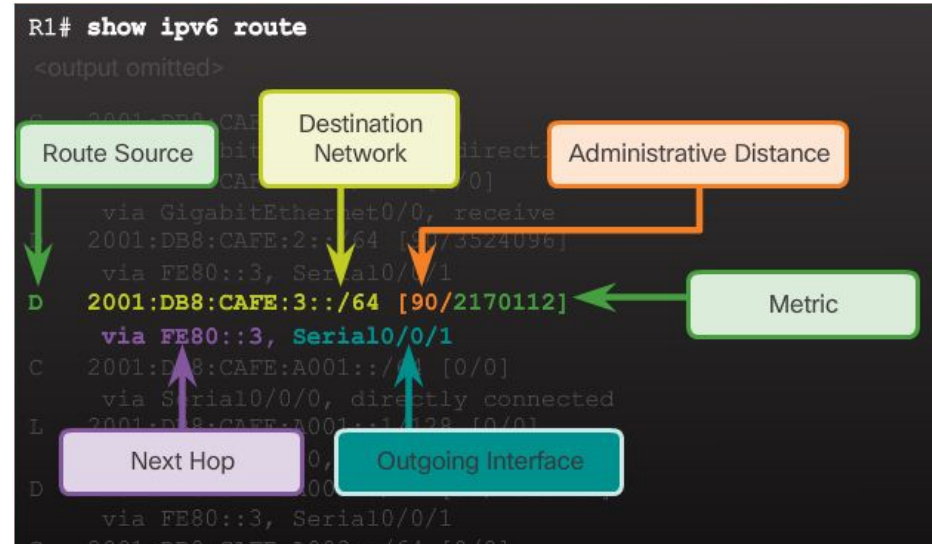


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- The remote IPv6 route entries also include the following:
 - Route source – How the route was learned. Common codes include O (OSPF), D (EIGRP), R (RIP), and S (Static route).
 - Next hop - Identifies the IPv6 address of the next router to forward the packet to.

The IPv6 router lookup process:

- Examines level 1 network routes for the best match.
- Longest match is the best match.





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